# MTH2005, modelling: theory and practice
# Assessed Case study 1

Deadline: noon on **Monday 8 December 2025**. Submit a LaTeX report of maximum 20 pages, taking care to explain your methodology as well as submitting your solutions and solution codes. **You are required to use python for the numerical solutions and to follow closely the methods defined in the MTH2005 lecture notes**. Where you have used python, clearly comment the commands used. Please also consult the *MTH2005 Guidance on coursework assessment and report writing* and the *GenAI guidance* in the Assessment panel on ELE. The total marks given for each question define the approximate proportions of the marks to be allocated.

1. **Fixed-point method**

   Solve

   $$f(x) = (x+5)^{1/3} - x + 3 = 0$$

   in the interval $x \in [3, 7]$ using the fixed-point method, rearranging the problem as $x = g(x)$. Consider the following two choices for $g(x)$:

   (a) $g(x) = (x+5)^{1/3} + 3$,

   (b) $g(x) = (x-3)^3 - 5$.

   Start the iteration with $x_0 = 6$. Using both the theory in the lectures and diagrams, show whether or not these choices converge. For any of the choices that converge, apply the fixed-point iteration 10 times to get a root. Also, verify that the root has been found, within a numerical error. Plot on the same graph: the function $f$, the two choices for $g$, the line $y = x$, and the location of the root. For each case, plot the stages of iteration with lines and arrows (over $g$ and $y = x$).

   **[30]**

2. **2D Newton Raphson**

   The Newton Raphson method is not only useful for finding the roots of functions, but can also be used to find the stationary points of functions. Consider the 2D function $F_{\text{main}}(x, y)$,

   $$F_{\text{main}}(x, y) = 3x^3 + 2y^3 + 2x^2 + 2xy + 2y^2 - 6x - 6y,$$

   where the notation $F_{\text{main}}$ for this scalar function is used to distinguish it from the vector function $\mathbf{F}$ used in the lecture notes for the 2D Newton-Raphson method.

In 2D, the vector gradient of $F_{\text{main}}(x, y)$ is $\nabla F_{\text{main}}$ and the stationary points are defined as

$$\nabla F_{\text{main}} = \begin{pmatrix} 9x^2 + 4x + 2y - 6 \\ 6y^2 + 2x + 4y - 6 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

giving two separate functions for the $x$ and $y$ components. In order to find a stationary point of $F_{\text{main}}$, the root of these two functions needs to be determined simultaneously.

Starting at the point

$$\mathbf{x}_0 = \begin{pmatrix} -5 \\ -1 \end{pmatrix},$$

take 10 iterations of the 2D Newton Raphson method to find a stationary point of $F_{\text{main}}(x, y)$. Using your knowledge of the matrices chapter of the module, identify the nature of this stationary point.

**[30]**

3. **Boundary-value matrix problem**

Consider the boundary-value problem

$$\frac{d^2y}{dx^2} = \cos x,$$

on a domain $[0, 2\pi]$ with a 'Neumann' boundary condition (prescribing the first derivative) on the left boundary and a Dirichlet one (prescribing $y$) on the right where

$$y'(0) = y(2\pi) = 0.$$

Calculate the true solution. Modifying the boundary-value problem in the lecture notes, form a matrix inversion problem to solve this equation. For the boundary conditions, you can use a forward Euler approximation, for example at the left boundary use

$$y'(0) \simeq \frac{y_1 - y_0}{\Delta x}$$

Modify the code **boundary_value.py** and solve the matrix problem numerically for a range of total number of points between 4 and 100. Calculate the error $\epsilon$ in the solution

$$\epsilon = \overline{|y_{num} - y_{true}|}$$

where the over-bar indicates an average over all horizontal points. Here, $y_{num}$ and $y_{true}$ are the numerical and true values respectively. Plot the error against grid length on a log-log plot. Thus infer the power law for the convergence of the error

$$\epsilon \sim \Delta x^n$$

where $n$ is an integer. Use your knowledge from the ODEs course to explain this rate of convergence.

**[40]**

**TOTAL 100 marks**